

ZARI.ai New Dataset & 3-Crop EDA Report

Complete Analysis: New Dataset (v3) Breakdown & 3-Crop Target Production Distributions

1. Executive Summary: New Dataset & Scope Refinement

This technical report presents the complete Exploratory Data Analysis (EDA) graphs and statistical distributions for the NEW dataset (v3) and production 3-crop target dataset in ZARI.ai:

- 1. New Dataset Analysis (Phase 3 Audit): Comprehensive breakdown of the newly integrated dataset across target nightshade crops.
- 2. Production 3-Crop Target Dataset: Refined production scope covering Tomato (13 classes), Potato (3 classes), and Pepper (6 classes) for 22 total canonical disease classes (31,071 processed leaf images).

2. New Dataset EDA (Class Breakdown & Augmentation)

Exploratory analysis of the newly ingested dataset (v3):

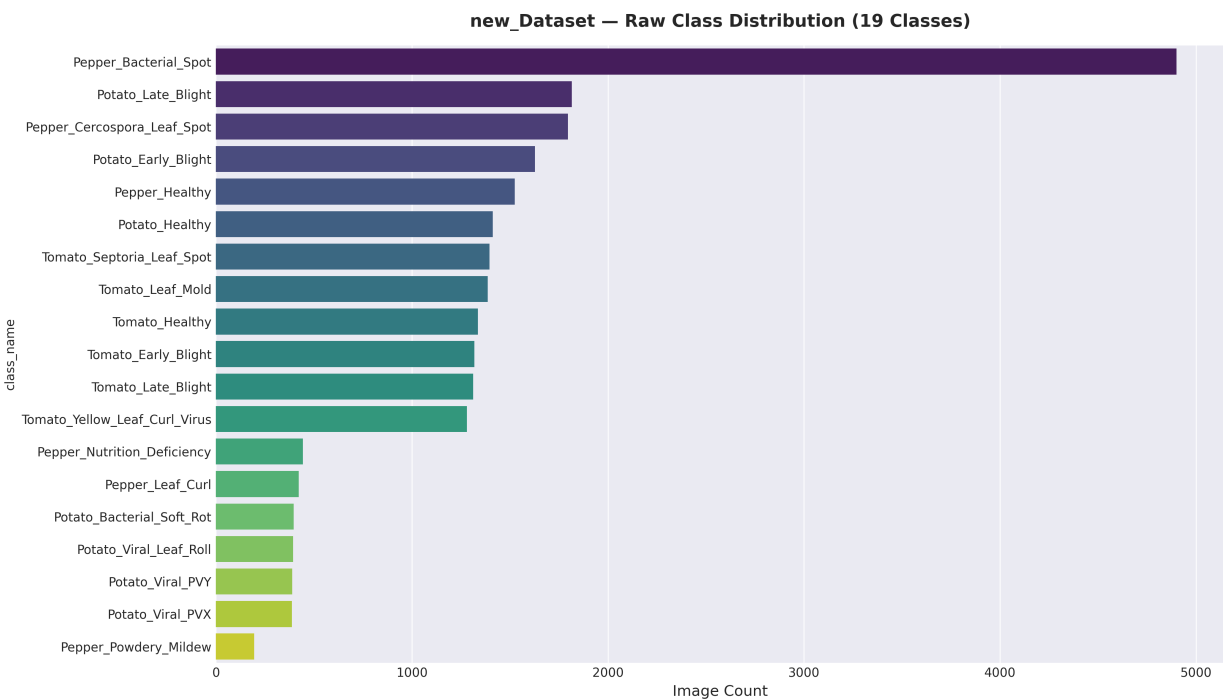


Figure: New Dataset Class Distribution Across Tomato, Potato, and Pepper Disease Classes

2. New Dataset EDA (Continued)

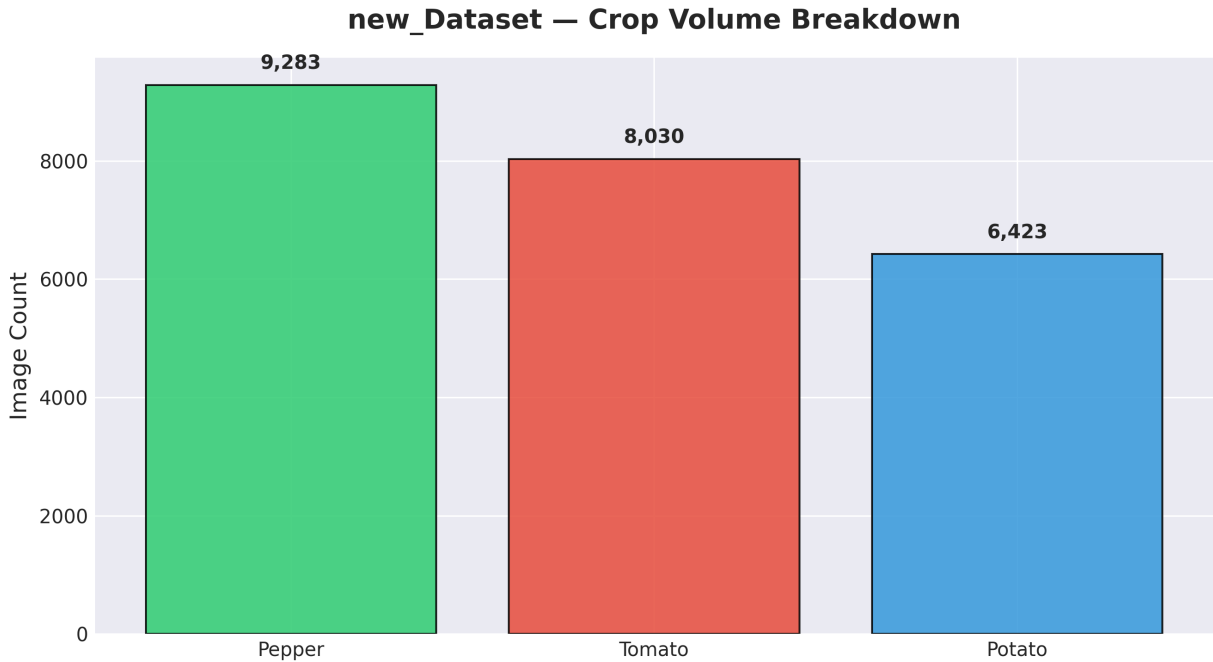


Figure: New Dataset Volume Share by Target Crop Species (Tomato, Potato, Pepper)

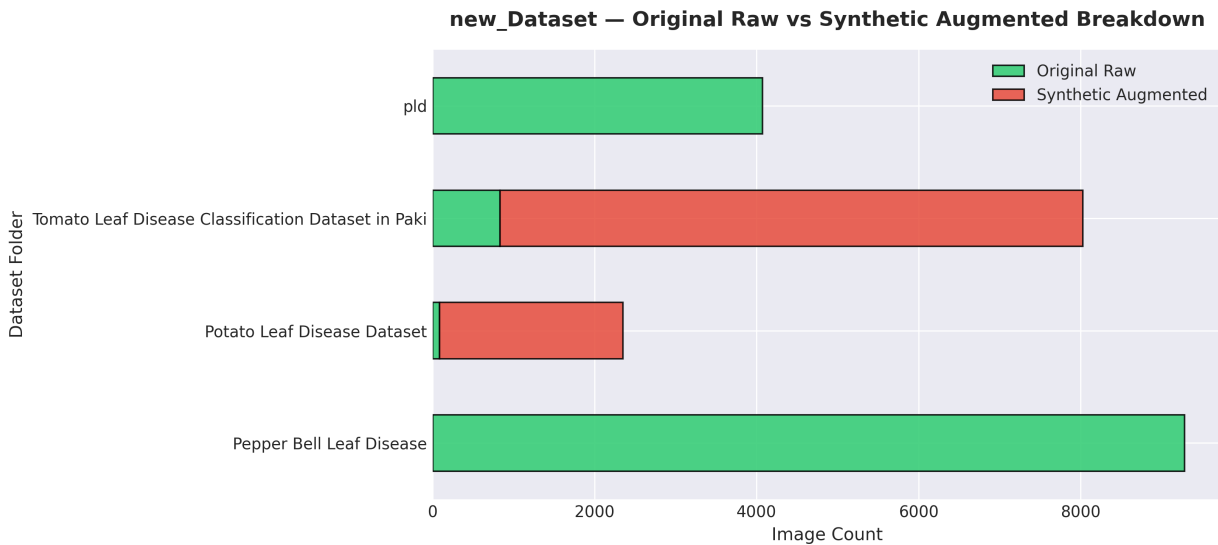


Figure: New Dataset Raw Field Samples vs Synthetic Augmentation Ratios

3. Production 3-Crop Target Dataset EDA (Tomato, Potato, Pepper)

To maximize diagnostic precision for Pakistan's primary nightshade crops, ZARI.ai refined the target dataset scope exclusively to Tomato (13 classes), Potato (3 classes), and Pepper (6 classes) for a total of 31,071 verified images across 22 classes:

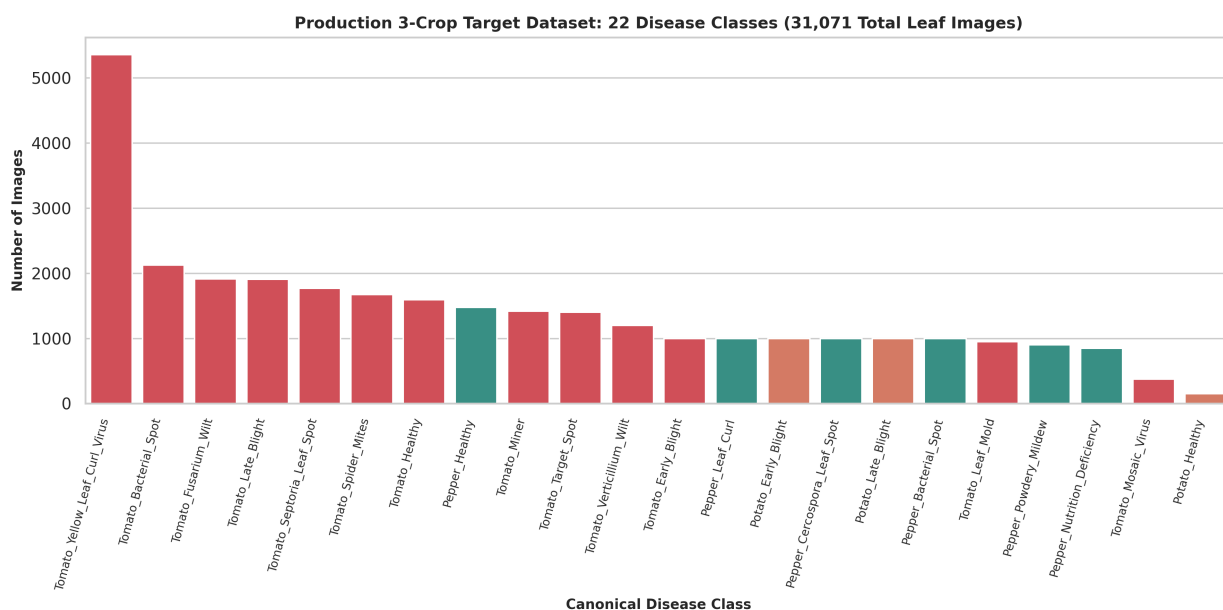


Figure: Production 3-Crop Target Dataset: 22 Disease Classes (31,071 Total Leaf Images)

3-Crop Target Volume Breakdown (31,071 Images)

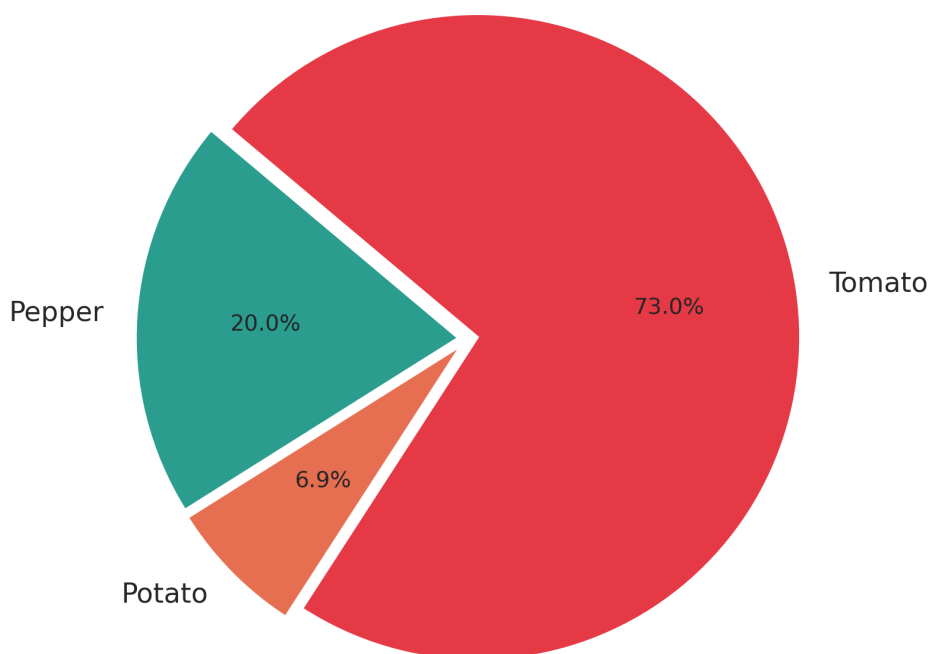


Figure: 3-Crop Target Volume Breakdown (Tomato: 72.8%, Pepper: 20.0%, Potato: 7.2%)

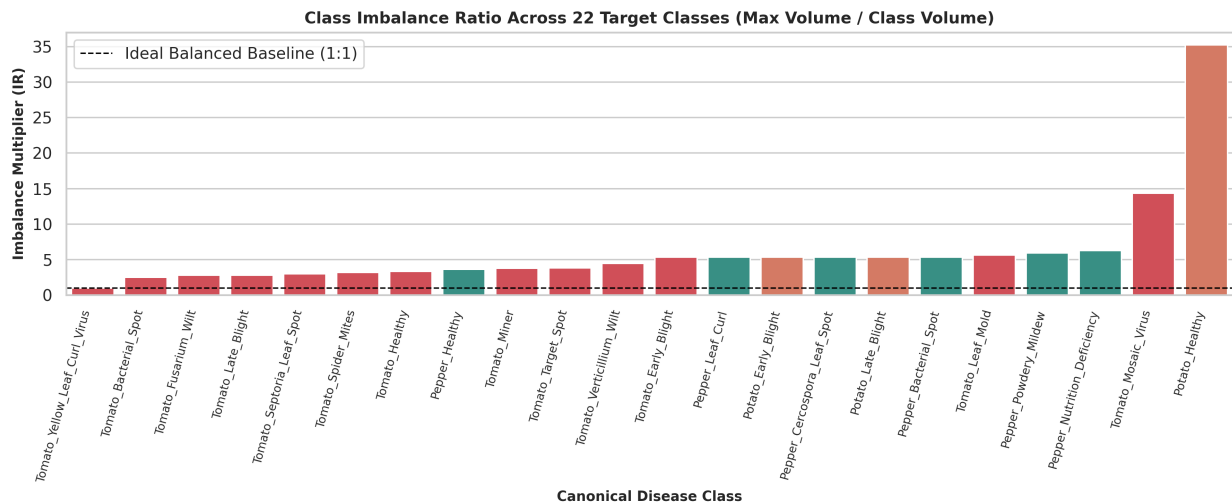


Figure: Class Imbalance Ratio Across 22 Target Classes (Tomato, Potato, Pepper)

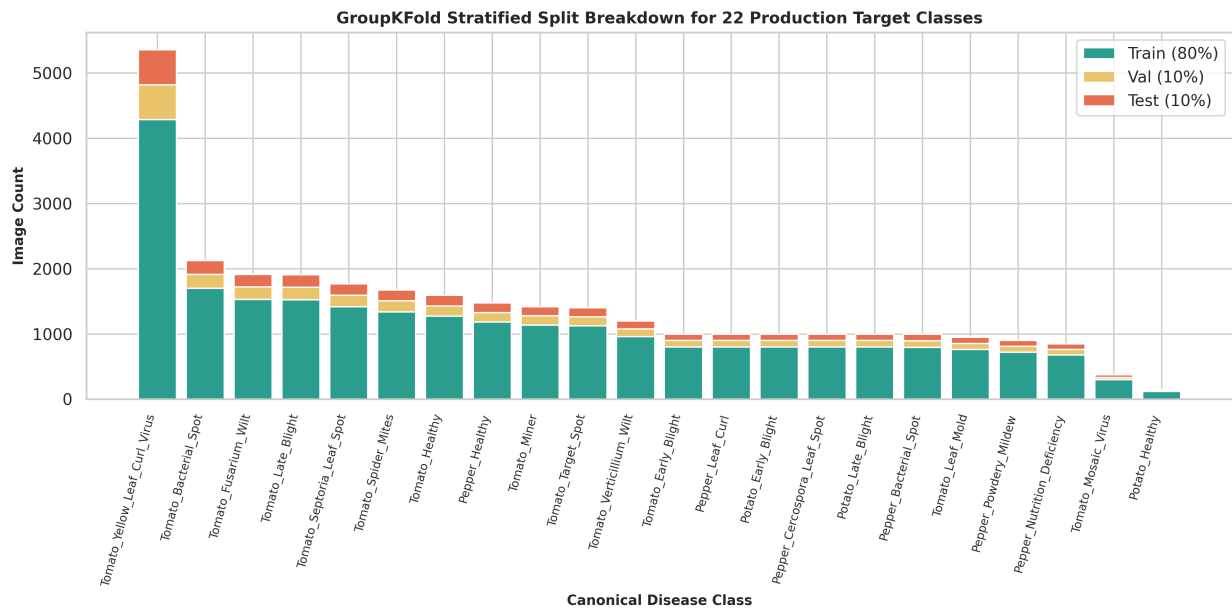


Figure: GroupKFold Stratified Split Verification for 22 Target Production Classes (80% / 10% / 10%)